

The Problem

In a catalytic-wall Taylor-Couette reactor, liquid mixed with reactant of concentration C_{in} enters the chamber in between an inner cylinder spinning at a rate ω and a stationary cylindrical shell. That cylindrical shell is a catalyst that triggers a reaction we assume to be infinitely fast.

So $C_{wall} = 0$ and conversion is limited only by mass transfer to the wall.

Finally what exits is our C_{out} .

What is the best way to spin the inner cylinder to get the most conversion X for the least motor power?

$$X = 1 - \frac{C_{out}}{C_{in}}$$

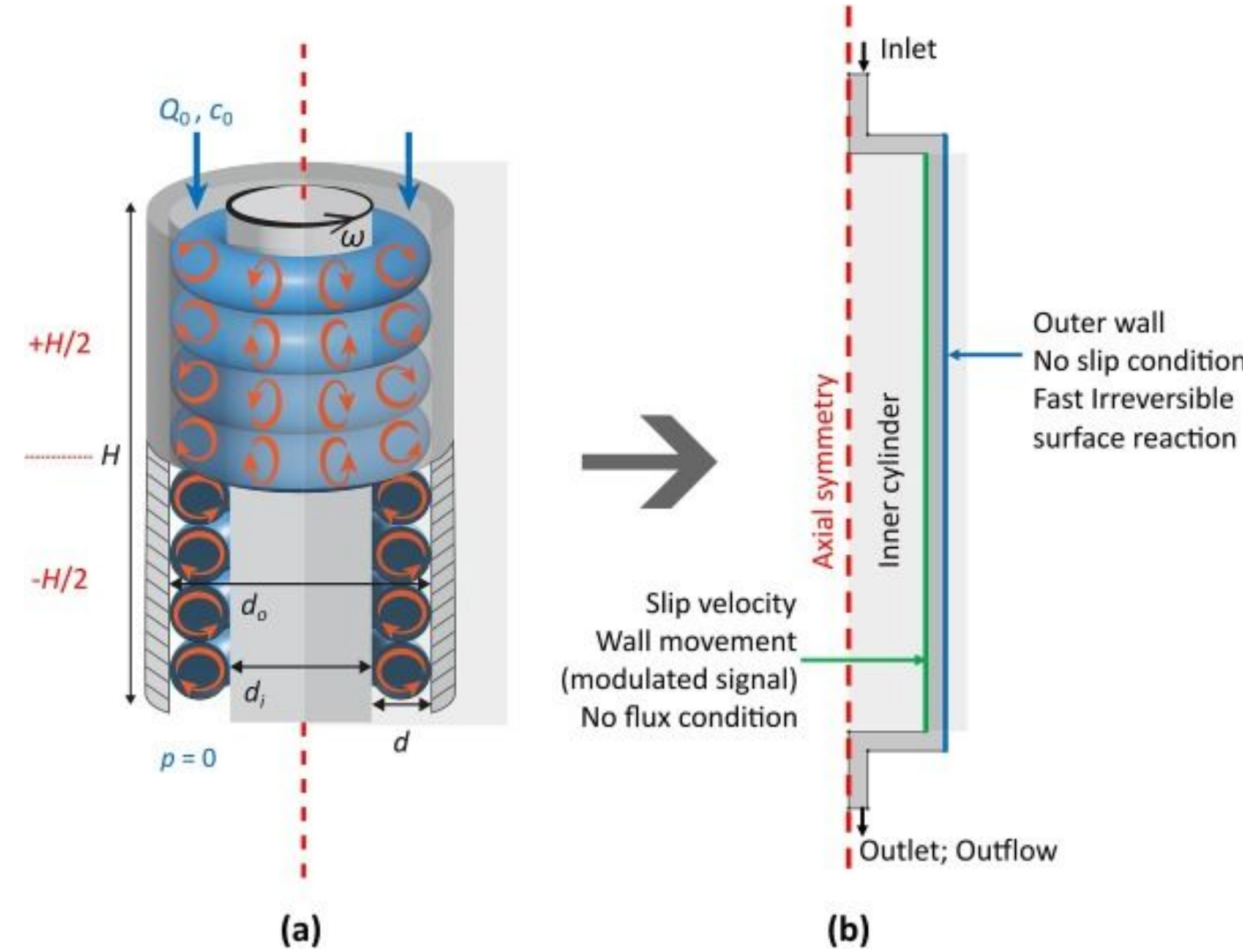


Figure from López-Guajardo et al. [1].

Our Method

1. OpenFOAM is an open source CFD software [3] and we use their PimpleFoam solver for:

Laminar incompressible flow with constant viscosity and no turbulence model

$$\nabla \cdot \mathbf{u} = 0$$

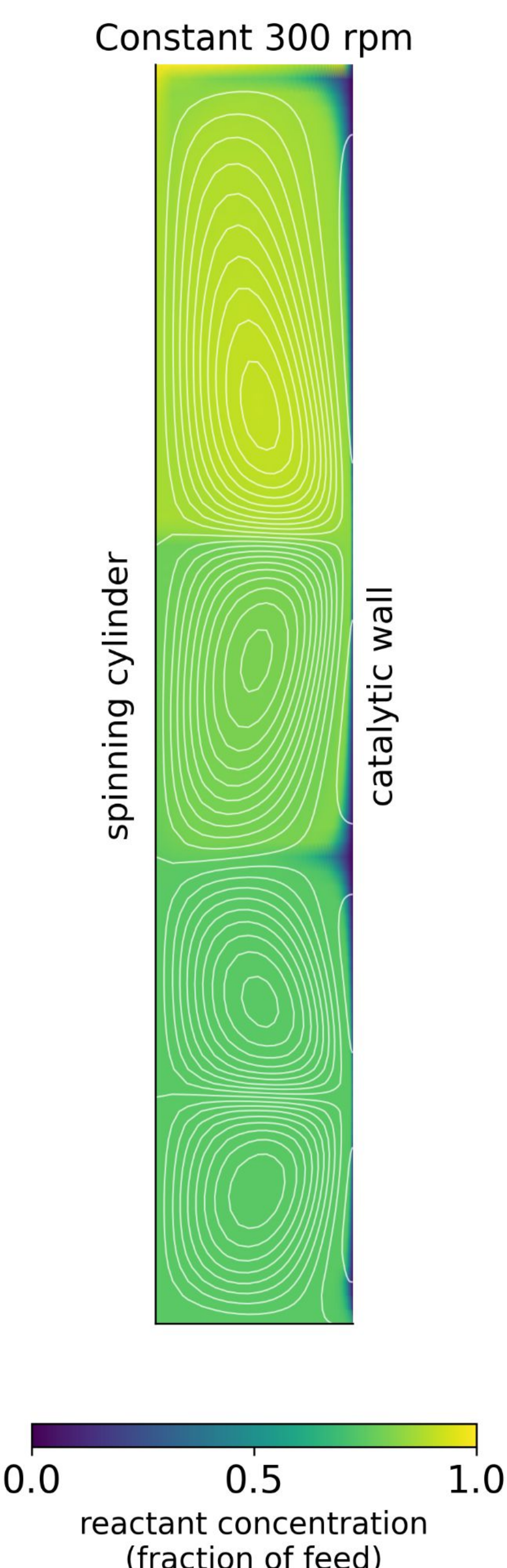
$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}$$

Scalar-Transport Equation

$$\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = D \nabla^2 c$$

Finite-volume, 2D axisymmetric, 3330 cells graded to 25 micrometers at the catalytic wall.

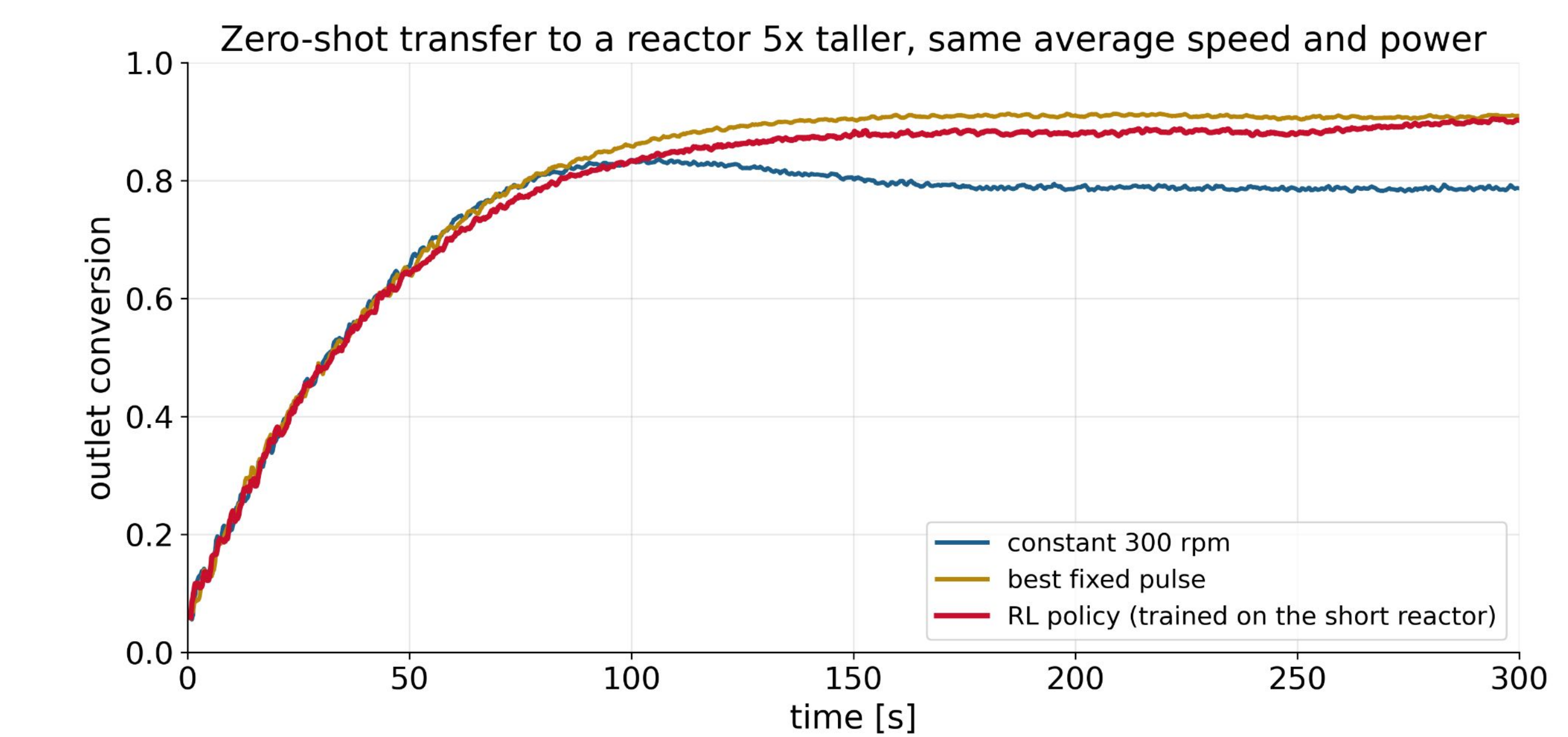
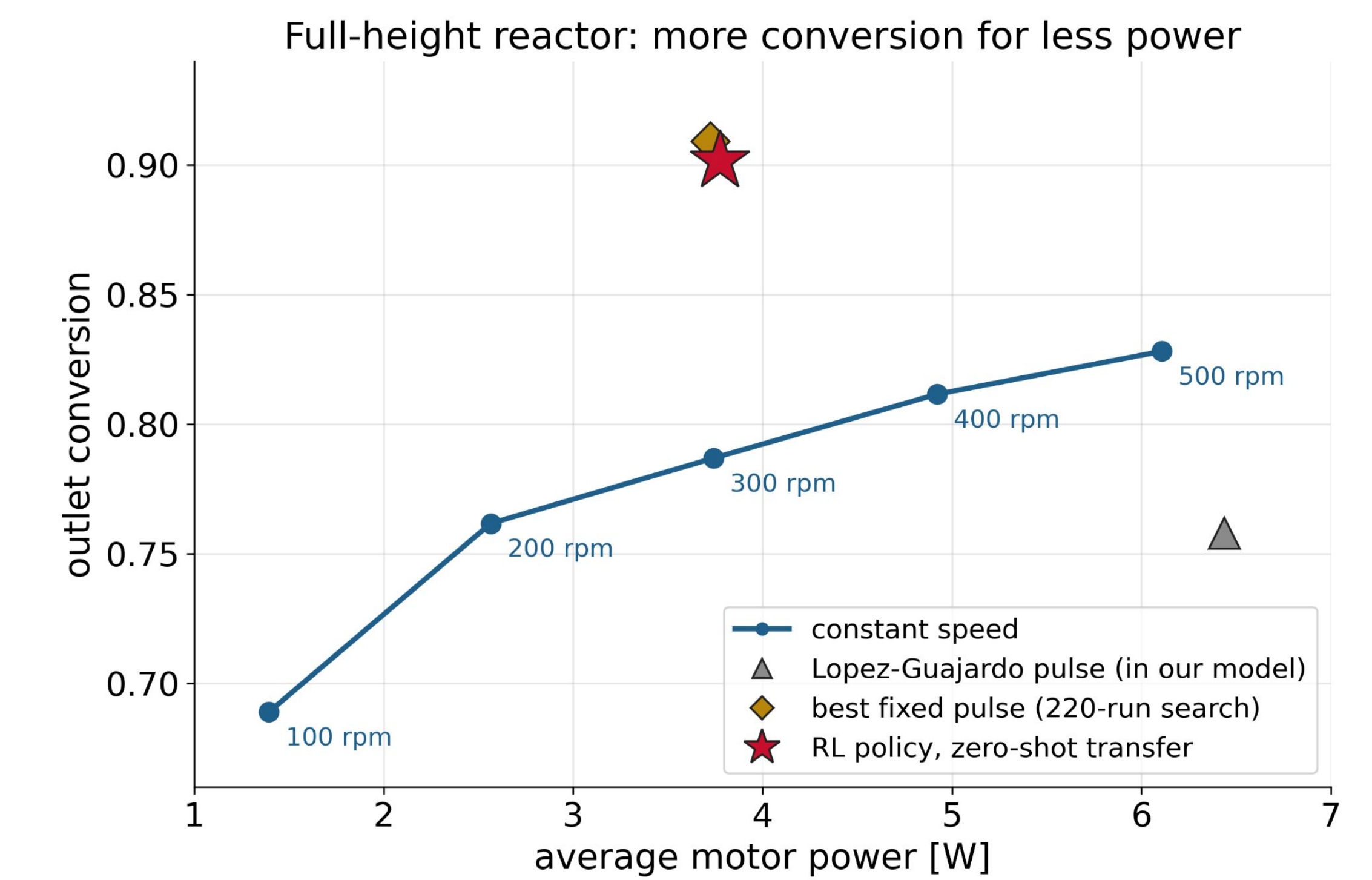
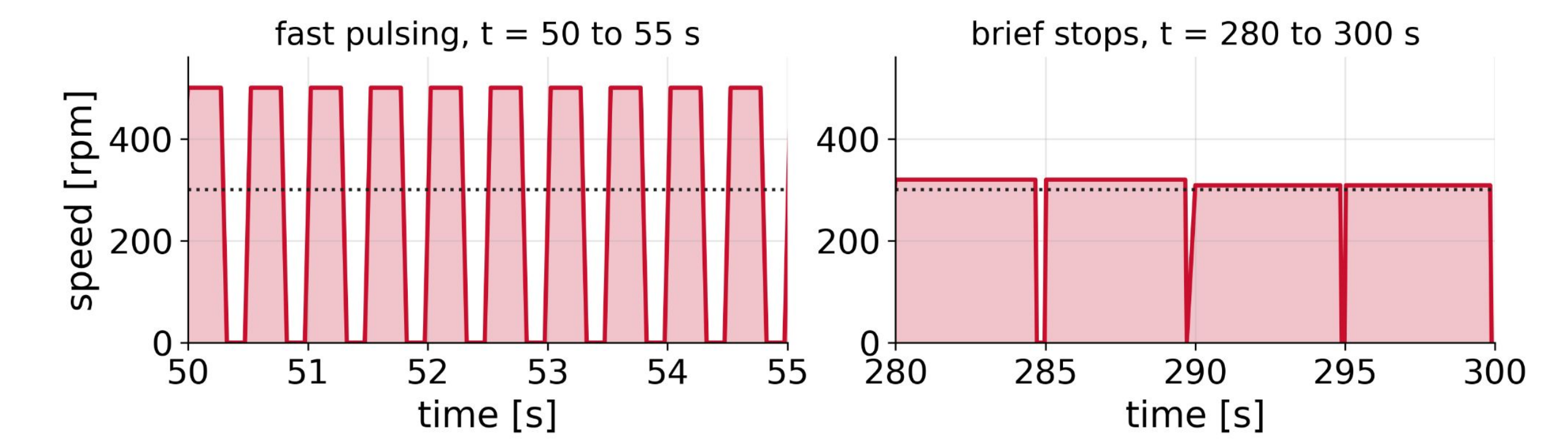
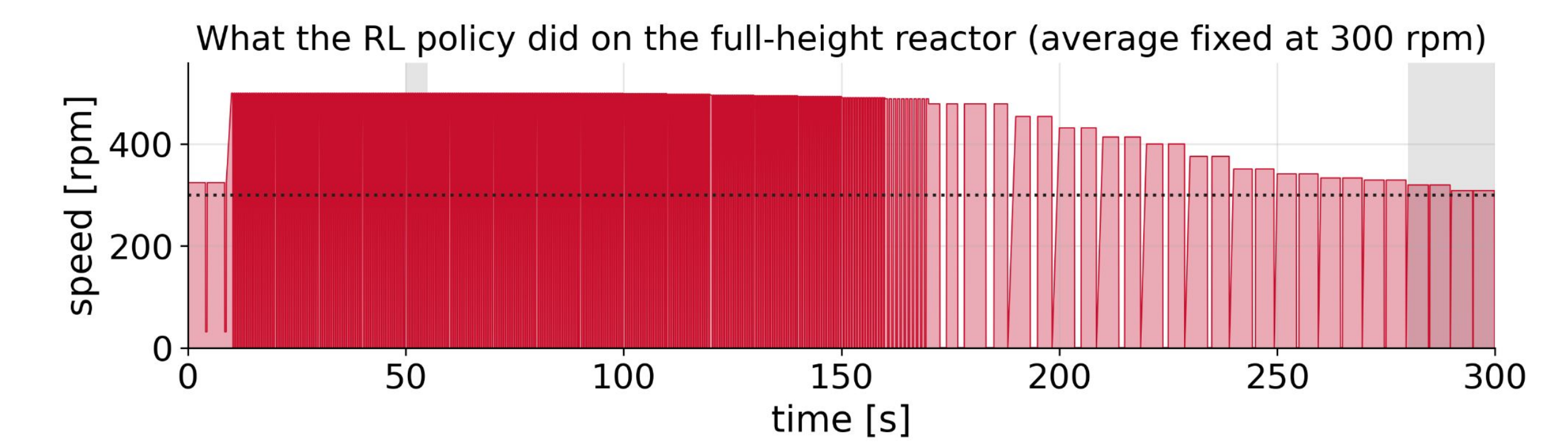
2. Gymnasium [4] creates an environment that allows our RL agent to communicate with our OpenFOAM simulation.
3. We chose TD3 [2] as our RL algorithm.
4. Every 10 seconds, the agent chooses the next pulse: its duty, its period, and the speed during the low part. The high speed is then set so the average is always 300 rpm. Its reward is conversion minus motor power. We trained on a reactor $\frac{1}{5}$ the full height, then tested on the full height.



Isaiah Bascom¹, Min Wang¹, Yuhe Wang²
¹ Department of Mathematics, University of Houston
² Institute for Scientific Computation, Texas A&M University

Reinforcement learning discovered a better way to run a catalytic-wall Taylor-Couette reactor

Brief stops are better than constant operation



Next Steps

1. An operator learning surrogate of the reactor dynamics would save us time in comparison to our OpenFOAM simulation.
2. We don't yet know why half-second stops are the best or why it chooses these patterns for certain. We estimate the vortices die in about 0.4 seconds and that the depleted concentration boundary layer reforms in about 0.8 seconds, so there may be a connection with the viscosity.
3. We would like to train RL under varying conditions such as the feed rate or viscosity.
4. Training in 3D or the real fluid's Schmidt number would help our simulations be more realistic and eliminate our biggest limitations.

References

- [1] E. A. López-Guajardo, R. Galluzzi, F. Delgado-Licona, R. Morales-Menendez, "Process intensification of a catalytic-wall Taylor-Couette reactor through unconventional modulation of its angular speed," Chemical Engineering Journal 489, 151174 (2024). doi:10.1016/j.cej.2024.151174
- [2] S. Fujimoto, H. van Hoof, D. Meger, "Addressing function approximation error in actor-critic methods," Proceedings of the 35th International Conference on Machine Learning (ICML), PMLR 80, 1587–1596 (2018).
- [3] OpenFOAM v2506, OpenCFD Ltd., www.openfoam.com (2025).
- [4] M. Towers et al., "Gymnasium: A standardized interface for reinforcement learning environments," arXiv:2407.17032 (2024).

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